OLYMPIAD · EVERYDAY MATHEMATICS · OLYMPIAD PREPARATION · SHORT NOTES

Chapter 13: Ratio and Proportion

Topic-wise short notes with solved examples

Comparing quantities by ratio, sharing in a ratio, proportion, and direct and inverse variation in everyday problems.

1 Ratio

- The **ratio** $a : b$ compares two quantities of the same kind. It is the fraction $\frac{a}{b}$ and has no units.
- A ratio is simplified by dividing both terms by their common factor: $45 : 60 = 3 : 4$.
- To **share** a quantity in the ratio $a : b : c$, add the parts $a + b + c$. One part is the quantity divided by this sum; then multiply by a, b and c.
- If $a : b$ and $b : c$ are given, make the value of b the same in both, then join them to get $a : b : c$.

Example 1.1 Divide Rs 360 in the ratio $2 : 3 : 4$.

SOLUTION

Total parts $= 2 + 3 + 4 = 9$. One part $= \frac{360}{9} = 40$.

Shares: $2 \times 40,\ 3 \times 40,\ 4 \times 40$.

Rs 80, Rs 120 and Rs 160

Example 1.2 If $a : b = 2 : 3$ and $b : c = 4 : 5$, find $a : b : c$.

SOLUTION

Make b equal in both: $a : b = 8 : 12$ and $b : c = 12 : 15$.

$8 : 12 : 15$

Example 1.3 The ages of two people are in the ratio $3 : 5$ and their sum is 40 years. Find the ages.

SOLUTION

One part $= \frac{40}{8} = 5$.

15 years and 25 years

2 Proportion

- Four numbers a, b, c, d are in **proportion** if $a : b = c : d$. Then $a \times d = b \times c$ (product of extremes = product of means).
- The **fourth proportional** to a, b, c is $d = \frac{bc}{a}$. The **mean proportional** between a and b is $\sqrt{ab}$.

Example 2.1 Find x if $4 : 7 = 12 : x$.

SOLUTION

$4x = 7 \times 12 = 84$.

$x = 21$

Example 2.2 Find the mean proportional between 4 and 9.

SOLUTION

$\sqrt{4 \times 9} = \sqrt{36}$.

6

3 Direct and inverse proportion

- If two quantities increase together at the same rate (cost and number of articles), they are in **direct proportion**: their ratio stays the same.
- If one quantity increases when the other decreases so that the **product** stays the same (workers and days), they are in **inverse proportion**.

Example 3.1 The cost of 5 pens is Rs 40. Find the cost of 12 pens.

SOLUTION

Cost of one pen $= 8$, so 12 pens cost 12×8.

Rs 96

Example 3.2 6 workers can finish a job in 12 days. How many days will 9 workers take?

SOLUTION

The work is $6 \times 12 = 72$ worker-days. Days $= \frac{72}{9}$.

8 days

REMEMBER

Before you calculate, decide in words: "more workers, fewer days" means inverse. "More pens, more cost" means direct.